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The Impact of Measurement Uncertainty on Hydrocarbon Allocation: Case Study for Upgrading Allocation Meters to Custody Class

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Abstract

Objective / Scope: This paper investigates the impact of flow measurement uncertainty on hydrocarbon production allocation accuracy in upstream oil and gas operations. It focuses on a real world case study where allocation grade meters were upgraded to custody transfer class in a multi-asset environment. The objective is to quantify how reduced measurement uncertainty influences allocation equity, allocated volumes to wells and reservoirs, Production Optimization and reservoir management decisions.

Methods, Procedures, Process: A systematic approach was used to assess the impact of measurement uncertainty, including uncertainty quantification, meter performance benchmarking, and allocation sensitivity analysis. Actual field data from several metering locations were collected before and after the upgrade to custody-class meters. Allocation simulations were run using different uncertainty inputs to evaluate resulting volume and value deviations. Calibration records and metering configurations were analyzed to determine typical sources of error. The analysis focused on how small variations in meter accuracy could lead to significant differences in production attribution reconciliation across joint venture partners and operational boundaries.

Results, Observations, Conclusions: The results revealed that upgrading to custody class meters reduced overall measurement uncertainty by up to 50%, leading to substantial improvements in allocation factors. In high volume assets, allocation discrepancies exceeding the acceptable levels, when comparing legacy meter data against the upgraded custody class instruments. These variances translated to material financial exposure, particularly in shared facilities with multiple equity partners. Sensitivity analyses performed demonstrated that a 1.2 % uncertainty improvement could result in millions of dollars in value reallocation annually.

Uncovered latent issues such as inadequate calibration frequency and data drift in legacy meters. These findings emphasized the importance of metrological integrity, especially in assets with tight contractual obligations.

In conclusion, the study confirms that meter upgrades coupled with digital allocation platforms can deliver significant technical, operational, and commercial benefits, while minimizing disputes and enhancing reporting confidence.

Novel/Additive Information: This paper is one of the few that quantitatively links field measurement uncertainty with allocation impact at a system level. It provides real world insights into how upgrading to custody grade meters can materially improve hydrocarbon accounting accuracy. It also introduces a replicable methodology for assessing value at risk due to measurement errors in multi-stakeholder oil and gas operations.

Introduction

This paper investigates the impact of flow measurement uncertainty on hydrocarbon production allocation accuracy in upstream oil and gas operations. It focuses on a real-world case study where allocation grade meters were upgraded to custody transfer class in a multi-asset environment. The objective is to quantify how reduced measurement uncertainty influences allocation equity, allocation volumes to wells and reservoirs, Production Optimization and reservoir management decisions.

The Company has embarked on a strategic EPC upgrade initiative transitioning from allocation-grade meters to custody and fiscal class metering systems across its upstream oil and gas assets, details as shown in Figure 1. This transformation is driven by the imperative to enhance hydrocarbon measurement accuracy, reduce allocation uncertainty, and align with international standards for fiscal accountability.

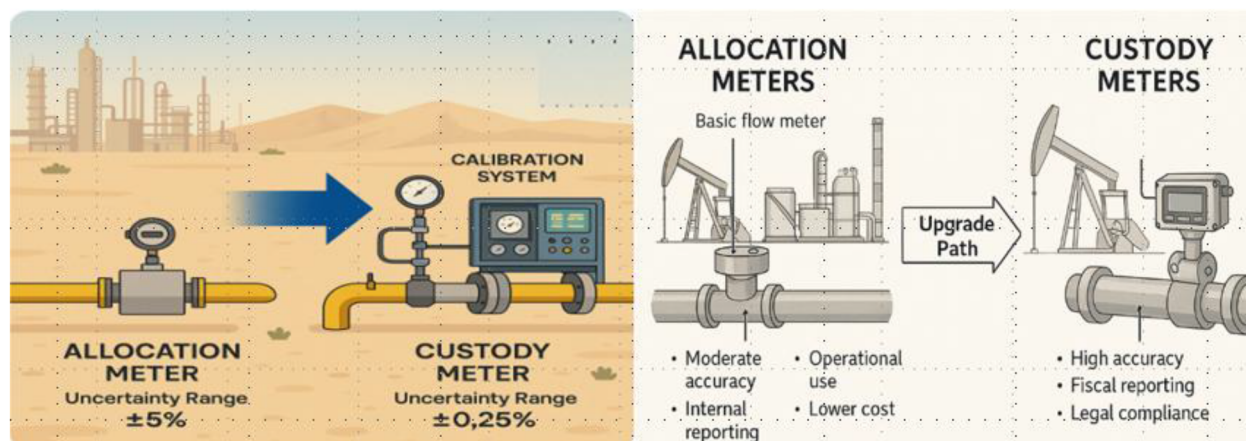


Figure 1—Upgrade from Allocation Meters to Custody Transfer Meters

The upgrade involves replacing legacy allocation meters with high-precision custody transfer meters—primarily turbine and Coriolis types—calibrated to an accuracy of $\pm 0.25\%$ or better. This shift ensures that production volumes are measured with greater reliability, directly impacting allocation equity, reservoir management, and production optimization decisions. The initiative has already demonstrated measurable improvements in allocation accuracy and reduced volumetric discrepancies between field exports and terminal receipts.

The impact on measurement uncertainty is profound. Prior to the upgrade, the company observed daily oil production volume variances of up to 3% of the daily production attributed to metering inaccuracies, blending shrinkage, and tank-level conversion issues. Post-upgrade investigations revealed that uncorrected figures in the Production Measurement and Allocation System (PMAS) and inconsistent proving practices were major contributors to these variances.

A critical yet frequently overlooked issue involves human error in data entry—specifically, the input of uncorrected meter readings (based on line conditions) instead of corrected values (standard conditions) into

the Production Measurement and Allocation System (PMAS) as shown in Figure 2. Additionally, failure to input the accurate percentage of Basic Sediment and Water (BS&W) in the Custody Transfer Meter form can significantly impact the back-allocation process within the material balance equation. A limitation arises when field laboratories report water content to the nearest increment (e.g., 0.025 mL), which can significantly affect back allocation accuracy. In cases where the basic sediment and water (BS&W) content in a centrifuge bottle falls below 0.025% by volume, the Karl Fischer (KF) Coulometric method is the preferred technique for precise water determination. This method offers superior sensitivity and accuracy for low water content samples compared to alternatives such as volumetric KF titration. As demonstrated in Figures 2 and 3, such measurement discrepancies can lead to notable errors in production accounting and allocation integrity.

Daily Report Crude Export Metering System				
Skid ID	U1001	Start Date & Time:	8/3/2022 0:00:00 hrs	
Standard Conditions:	STREAM 01	End Date & Time:	8/4/2022 0:00:00 hrs	
Pressure	0	STREAM 02	0	Units
Temperature	15			bar(g)
Meter ID Number	FIT100111			Deg C
K Factor	10000.000000	FIT100112	10000.000000	pls/m3
Previous Day Flow Rates:				
Gross Uncorrected Flow rate(GUVOL)	37037.88	0.00		bb/d
Gross Standard Flow rate(GSVOL)	36319.92	0.00		bb/d
Net Standard Flow rate(NSVOL)	36319.92	0.00		bb/d
Previous Day Totalizer's:				
Gross Uncorrected Volume(GUVOL)	37365.06	0.00		bbl
Gross Standard Volume(GSVOL)	36578.24	0.00		bbl
Net Standard Volume(NSVOL)	36578.24	0.00		bbl
Previous Day Flow Weighted Averages:				
Temperature	44.86	0.00		Deg C
Pressure	71.40	0.00		bar(g)
Line Density	795.45	0.00		kg/m3
Standard Density	812.56	0.00		kg/m3
Previous Day Station Totalizer's:				
Gross Uncorrected Volume(GUVOL)	37365.06			bbl
Gross Standard Volume(GSVOL)	36578.24			bbl
Net Standard Volume(NSVOL)	36578.24			bbl
Remarks	Acknowledge by			

Figure 2—Uncorrected and Corrected Meter Figures

Oil Meter Reading		BOOSTER TOTAL METER	
Imperial			
Begin Date		End Date	Total Days 0
Trans. ID		Transporter	Ticket Number 30
☐ Transfer?	☐ Use API 11.1 - 2004?	☐ 4 Decimals?	☐ Rollover?
Obs. Gravity 38.38	Obs. Temperature 60.0	Corrected Gravity 38.38	
Line Temp. 60.0	Temp. Corr. Factor (CTL)	Volume Corr. Factor (CTPL)	
Meter Pressure	Pressure Corr. Factor (CPL) 1.00000	Meter 673136.00	
Open 0.00	Shrinkage 1.0000	Compressibility 1.0000	
Meter Corr. 1.0000	Meter Adj.	BS&W% 0.000	
Oil 673136.00	Water 0.00	Total 673136.00	
Remarks	Updated by xxxxxxxx		

Figure 3—Base Sediment and Water (%BSW) Entry in the PMAS Form

To mitigate these risks, the Company has implemented a rigorous verification and maintenance program. This includes scheduled calibration and proving cycles, centralized contract management, enhanced sampling protocols and automated data validation warnings to ensure data integrity and compliance.

The EPC upgrade from allocation-grade to custody and fiscal class metering marks a significant milestone in the company's pursuit of operational excellence. This transition enhances the precision of hydrocarbon

measurement and plays a critical role in protecting revenue integrity by substantially reducing uncertainty in production accounting.

Methods, Procedures, Process

A systematic approach was used to assess the impact of measurement uncertainty, including uncertainty quantification, meter performance benchmarking, and allocation sensitivity analysis. Actual field data from several metering locations were collected before and after the upgrade to custody-class meters. Allocation simulations were run using different uncertainty inputs to evaluate resulting volume and value deviations. Calibration records and metering configurations were analyzed to determine typical sources of error. The analysis focused on how small variations in meter accuracy could lead to significant differences in production attribution reconciliation across joint venture partners and operational boundaries.

The following assumptions and references were made before, during and after the assessment of the impact of the measurement:

- **Uncertainty and Accuracy Compliance:** The Company requires custody meters to maintain an overall combined standard uncertainty of $\pm 0.25\%$ or better, calculated in accordance with ISO 5168 and API MPMS Chapter 13.3 using the root-sum-square method for individual uncertainty components.
- **Meter Factor and Repeatability Standards:** The acceptable meter factor—defined as prover volume divided by meter volume—should fall within 0.9900 to 1.0100 (average uncertainty of 0.1% to 0.2% and not more), with repeatability between 0.01% and 0.05%, striving to achieve a repeatability of 0.02% or better.
- **Proving Conditions and Run Requirements:** Custody meter proving must be conducted under stable operating conditions—consistent flow rate, pressure, and temperature—with a minimum of five consecutive runs to ensure repeatability and accuracy, as per API MPMS Chapters 4 and 12.2.3.
- **Calibration and Traceability Standards:** Proving systems, typically Small Volume Provers (SVP) or Bi-Directional Provers, must be calibrated with corrections for temperature, pressure, and density, and traceable to National Institute of Standards and Technology (NIST) certified measures.
- **Frequency of Proving and Calibration:** The Company requires custody meter proving every six months, annual calibration for SVPs and tank portable provers, and triennial calibration for bi-directional and compact uni-directional provers.
- **Manual Sampling Standards:** The Company follows API MPMS Chapter 8.1 and ASTM D4057 for manual sampling of petroleum and water content, detailing procedures for representative sample collection, container handling, and preparation.
- **Automatic Sampling Standards:** API MPMS Chapter 8.2 governs automatic sampling practices, emphasizing system design, sample integrity, and calibration protocols to ensure accurate and representative sampling.
- **Laboratory Analysis for Low Water Content:** For BS&W levels below 0.025% by volume or water content reported to the nearest increment (e.g., 0.025 mL), the Karl Fischer Coulometric method is recommended due to its superior sensitivity and precision, minimizing allocation errors caused by rounding limitations in field laboratory centrifuge bottle / tube reporting.

Uncertainty in measurement affects financial reporting, partner trust, and compliance. Allocation meters are often not accurate enough for financial allocation. Uncertainty may be defined using four critical indicators such as: K-Factor, Meter factor, Linearity and Repeatability.

The K-Factor of a Coriolis or Turbine meter represents the number of pulse outputs generated per unit of mass or volume passing through the meter. For pulse-output meters, the K-Factor is typically expressed in units of pulses per unit volume or mass.

This value is determined by passing a known quantity of fluid—either by mass or volume—through the meter while recording the total number of pulses using a counter. The K-Factor is then calculated as:

K-Factor = Total Pulse Count / Known Volume or Mass of Fluid

K-factor = (Total pulses) / (Unit mass or volume).

The Meter Factor is a correction coefficient derived during meter proving and is commonly used to adjust the indicated flow rate of a meter. By multiplying the meter's measured flow by the calculated meter factor, a more accurate representation of the actual volume is obtained for inventory and accounting purposes. The meter factor is determined using the following formula:

Meter Factor = Prover Volume / Measured Volume (or Mass)

This ensures that any discrepancies between the meter reading and the actual volume passed through the prover are accounted for, thereby enhancing measurement accuracy

Linearity in Flow Measurement. The ideal accuracy curve for volumetric flow meters is a straight line, indicating a consistent meter factor across the operating range. Meter linearity refers to the total deviation of the actual accuracy curve from this ideal straight line, measured between the minimum and maximum recommended flow rates.

Linearity is expressed as a percentage and calculated using the following formula:

Linearity (%) = [(Max Meter Factor – Min Meter Factor) / Mean Meter Factor] × 100

This metric is essential for evaluating the meter's performance consistency and its suitability for precise flow measurement applications.

Repeatability refers to a meter's ability to consistently produce the same reading under identical flow conditions. It is assessed by analyzing the results of two or more proving runs. Repeatability is expressed as a percentage and calculated using the following formula:

Repeatability (%) = [(Max Meter Factor – Min Meter Factor) / (Min Meter Factor)] × 100

This metric is essential for evaluating the reliability and stability of a meter's performance over repeated measurements.

The Company adopts the ISO 5168-compliant uncertainty formula which recommends expressing the combined standard uncertainty as the square root of the sum of the squares of individual component uncertainties as per the below formula:

$$U_c = (U_K^2 + U_{MF}^2 + U_{lin}^2 + U_{rep}^2)^{0.5}$$

$$u_c = \sqrt{u_K^2 + u_{MF}^2 + u_{lin}^2 + u_{rep}^2}$$

Where:

- u_c = combined standard uncertainty
- u_K = uncertainty due to K-Factor
- u_{MF} = uncertainty due to Meter Factor
- U_{lin} = uncertainty due to Linearity
- u_{rep} = uncertainty due to Repeatability

This formula is based on the assumption that the uncertainties are independent and follow a normal distribution, in accordance with the principles outlined in ISO 5168. The Accuracy of any meter is the percentage cumulative combined uncertainty and is an indication of the combined standard uncertainty.

The graph in [Figure 4](#) below illustrates how key parameters—**K-Factor**, **Meter Factor**, **Linearity**, and **Repeatability**—were incorporated into the ISO 5168-compliant uncertainty formula to evaluate their impact on the accuracy of custody transfer meters, specifically within the ±0.25% tolerance range.

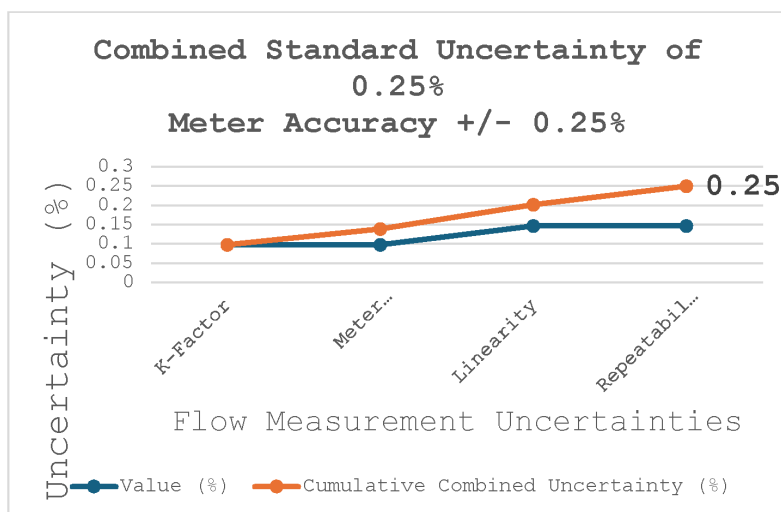


Figure 4—Combined Standard Uncertainty

Comparable evaluations were conducted for allocation meter accuracy levels of $\pm 5\%$, $\pm 0.5\%$, and $\pm 0.375\%$. A sensitivity analysis was then conducted across confidence intervals ranging from 95% to 99.75%, as depicted in Figure 5 below.

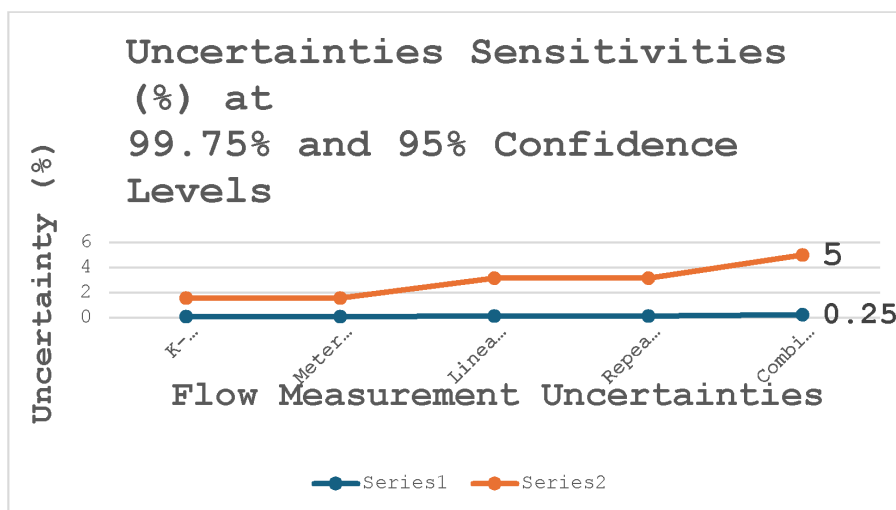


Figure 5—Uncertainties Sensitivities

To maintain meter integrity and minimize measurement uncertainty, any allocation meter that produces a meter factor deviating by more than 2.0% from its previous value during proving must be adjusted and successfully reproved before being returned to service.

When a meter is outside the acceptable meter factor tolerance—2.0% for allocation meters and 0.25% or more for custody meters—the volume recorded since the last valid proving must be corrected in accordance with applicable regulations, system agreements, or contractual terms.

If a proving results in a meter factor deviation equal to or exceeding the tolerance limit, the meter must be repaired and successfully reproved before resuming operation.

Custody Transfer Meters should be proved at intervals not exceeding six months to ensure continued accuracy. Fiscal Meters, on the other hand, must be proved at every offload or export operation to maintain measurement integrity:

- **Bi-directional Provers** are recommended for Fiscal Meter proving at **Export (External Boundary) Terminals**, whereas **Portable Provers** are suitable for Custody Transfer Meter proving at **Internal Boundary Nodes** and major oil pipelines transporting cross-field hydrocarbons within the same company.
- The proving setup generally employs either a Small Volume Prover (SVP) or a Bi-Directional Prover, and the calibration process must incorporate adjustments for temperature, pressure, and fluid density. The proving procedure must be carried out under stable flow conditions, with at least five proving runs required to ensure accuracy and reliability.
- **Prover Calibration** must be conducted annually. **Meter Calibration** is required following any major repair activity or if the meter fails to meet proving standards.
- A **complete performance curve** must accompany the calibration certificate. A new performance curve shall be generated under the following conditions:
 - Commissioning of a new line meter.
 - Overhaul or repair of an existing line meter.
 - Failure of the line meter to pass check runs.
 - Deviation of the meter factor from the established performance curve beyond acceptable limits.
 - Significant changes in operating conditions.
- **Custody transfer meters** are required to achieve an overall measurement accuracy of $\pm 0.25\%$ or better, with a repeatability threshold of 0.02% or less

Results, Observations, Conclusions

Sensitivity analysis comparing data from Allocation Meters before the upgrade and Custody Meters after the enhancement shows that volume variance between fields and terminals was notably higher—reaching up to 3% of daily production—when using Allocation Meters. In contrast, the variance with Custody Meters has dropped to a moving average of around 0.5%, reflecting an improvement exceeding 600%, as depicted in Figure 6 below

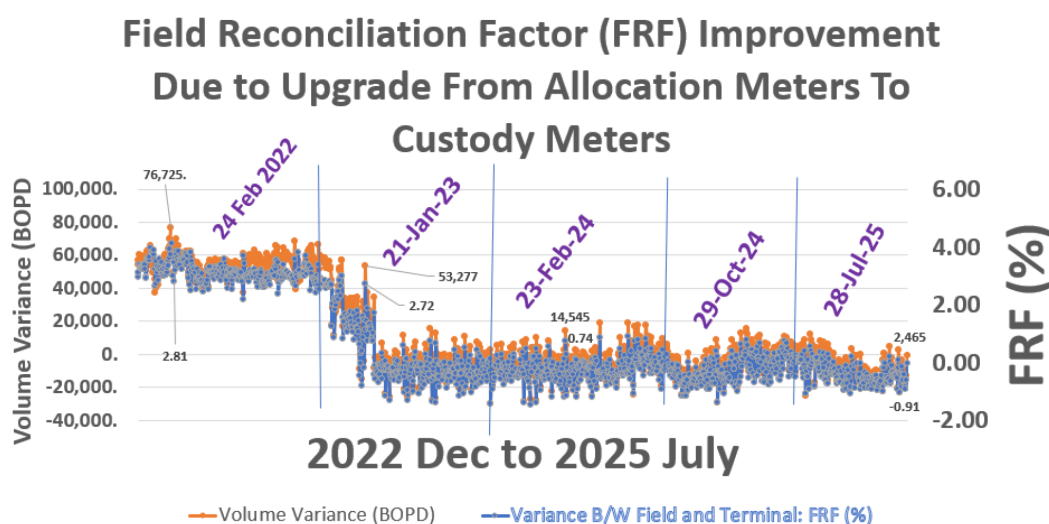


Figure 6—Reconciliation Factor Improvement due to Upgrade

The key performance indicators (KPIs) utilized are the Well Reconciliation Factor (WRF) and the Field Reconciliation Factor (FRF). The WRF is calculated as the ratio of the actual production at the field level to total theoretical production at the well level (based on well tests). The FRF is defined as the ratio of actual

production measured at the terminal to the total production recorded at the field level. The WRF measures the sum of continuously estimated well rates, which should fall within $\pm 5\%$ of the total field measured liquid rate and within $\pm 10\%$ for each individual phase, in accordance with the Company Back Allocation Guideline. The FRF, relevant to stabilized crude oil gathering networks, represents the hydrocarbon reconciliation factor at the field level, which should fluctuate randomly within $\pm 2\%$, maintaining a rolling mean between 0.98 and 1.0, also as per the guideline. These KPIs provide a clear assessment of meter performance across various allocation nodes at both field and terminal levels. The WRF highlights discrepancies between Custody and Allocation Meters, while the FRF reflects differences between Custody and Fiscal Meters. Notably, the abnormal variance between Allocation and Fiscal Meters was promptly resolved following the upgrade of Allocation Meters to Custody Meters. Once the FRF was recalculated using field custody transfer nodes against Terminal Fiscal Meters, a significant reduction in volume variance was observed, with the reconciliation factor dropping from 3% to a moving average of 0.5%, directly demonstrating the superior accuracy of Custody Meters over the previously used Allocation Meters.

Figure 7 below also illustrates the variance in daily production volumes between the fields and the actual volumes received at the terminals

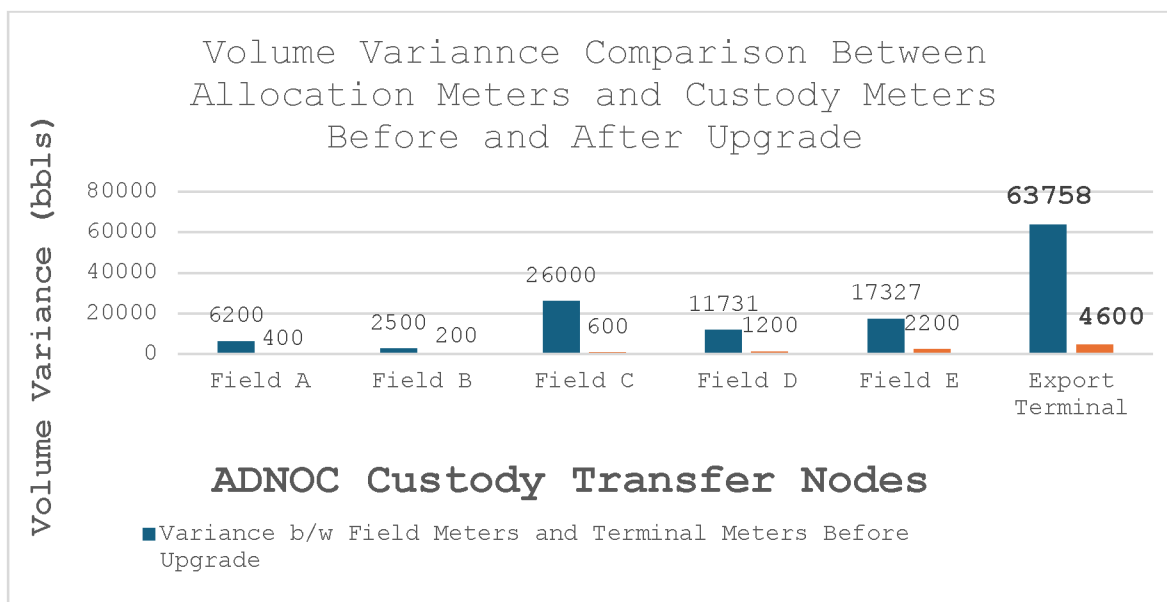


Figure 7

These discrepancies led to substantial financial exposure, especially for organizations with high daily production rates. Sensitivity analysis indicated that reducing uncertainty by just 1.2% could result in annual value reallocations worth several hundred thousand dollars, as demonstrated in Figure 8, the Meter Error versus Revenue Inconsistency graph below.

The Supporting calculations and Assumptions are as below:

1. Discount factor - 9%
2. Costs and savings incurred at the start of the year
3. Technology is adopted by peer companies
4. Efficiency is calculated at 95%, 90% and 70% for Best, Mean and Worst-case scenario respectively

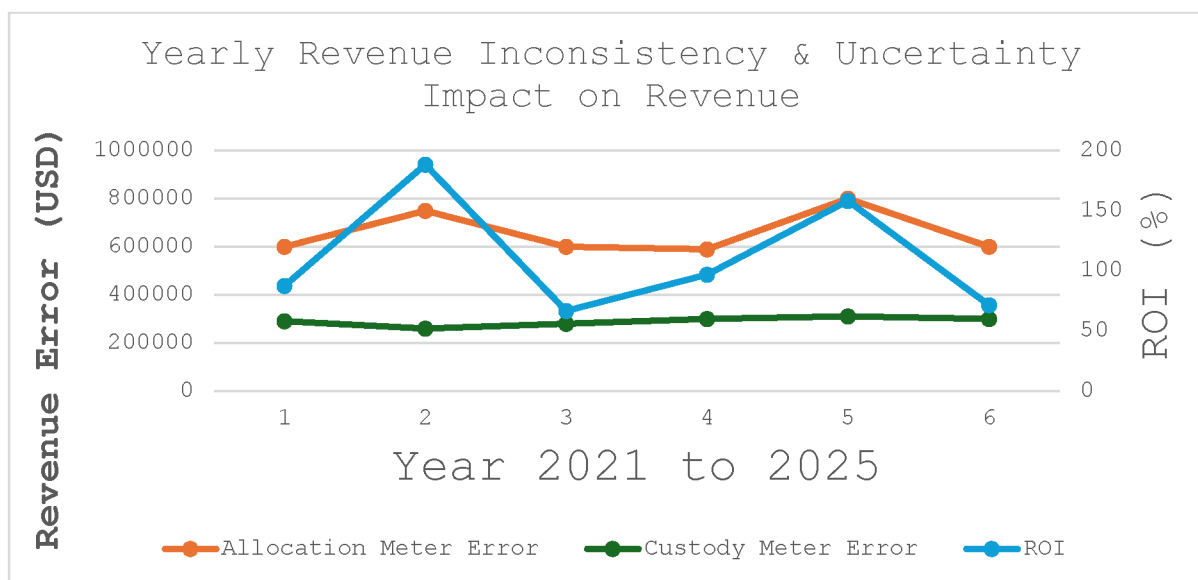


Figure 8—Meter Error versus Revenue Inconsistency

The analysis highlights the compelling financial case for upgrading allocation meters to custody-class standards. The initiative demonstrates a substantial return on investment, with ROI exceeding 100% within the first year of implementation. Even under conservative assumptions, the net present value (NPV) remains positive, confirming the robustness of the investment across varying operational scenarios.

A sensitivity analysis was conducted across three scenarios—Best Case (95%), Mean Case (90%), and Worst Case (70%)—evaluating both ROI and NPV. In all cases, total cash inflows significantly outpaced cash outflows, with ROI consistently achieved within one year. Notably, during the first two years alone, the cumulative cash inflows far exceeded the associated expenditures, reinforcing the financial viability of the upgrade strategy.

These findings underscore the strategic importance of metrological integrity in hydrocarbon allocation, especially in high-volume, multi-stakeholder environments. The upgrade not only enhances measurement accuracy but also delivers measurable commercial benefits, reduces disputes, and strengthens stakeholder confidence.

The deployment of custody transfer metering systems across field locations has markedly improved metering precision, reducing the uncertainty margin from $\pm 5\%$ to approximately $\pm 0.25\%$. This advancement has been consistently validated through recent facility capacity assessments and is reflected in daily and monthly allocation material balance factors.

All Company crude volumes from custody transfer nodes are consolidated at a central gathering hub prior to pipeline transfer to terminal and mega storage facilities. The associated Reconciliation Factor (RF), which accounts for volumetric shrinkage and other systemic losses, is uniformly applied across all custody transfer nodes. This RF is dynamic, influenced by real-time production rates and operational conditions, and has ranged from 0.9988 ($\pm 0.12\%$) to 0.988 ($\pm 1.2\%$) between late 2022 and early 2025.

To ensure robust design and operational reliability, a conservative design margin of $\pm 1.2\%$ has been incorporated into the Technical Rate calculations used for sizing surface infrastructure across all Asset Hubs and Central Degassing Station (CDS) facilities.

The implementation of First Tier Custody Transfer meters in satellite fields has led to a measurable improvement in the Reconciliation Factor (RF), with values ranging from 0.988 ($\pm 1.2\%$) to 0.980 ($\pm 2.0\%$). This variation is attributed to the presence of Allocation Meters within the satellite network, which introduces additional measurement uncertainty. Consequently, a conservative design margin of at least $+2\%$ was recommended for Technical Rate calculations when determining the maximum capacity thresholds of

surface facilities across all satellite assets. These sensitivity analyses were conducted strictly for process facility design capacity sizing purposes and not intended to support production increases.

A quantifiable enhancement of 0.6% in RF accuracy was observed, improving from 1.8% in 2022 during the initial upgrade phase to 1.2% by early 2025. This gain in metering precision has been incorporated into process facility capacity design considerations, enabling a potential uplift of 0.6% in Field Sustainable Oil Production Rate (FSOPR) and Field Sustainable Gas Production Rate (FSGPR) for assets previously operating with a 1.8% design margin.

This study affirms that upgrading metering infrastructures specifically transitioning from allocation-grade to custody and fiscal-class meters—when integrated with digital allocation platforms, yields substantial technical, operational, and commercial advantages. These enhancements not only reduce measurement uncertainty but also minimize disputes and significantly improve confidence in production reporting.

The transition from allocation-grade to custody and fiscal-class metering within the company has significantly reduced measurement uncertainty, resulting in transformative enhancements across hydrocarbon allocation and reservoir management workflows. With improved metering precision, hydrocarbon volumes are now attributed more accurately to their respective contributing assets, minimizing volumetric discrepancies and enabling a more equitable distribution of production entitlements among stakeholders. Additionally, the removal of systemic biases—previously caused by limitations in metering technology—has reinforced the integrity and reliability of the allocation process.

Accurate flow measurement is critical for reliable back-allocation of production volumes to individual wells and reservoirs, particularly in commingled systems where multiple streams converge at shared infrastructure. The improved granularity supports precise well-level performance monitoring and reservoir depletion analysis. With reduced uncertainty, production engineers can make more informed decisions regarding choke management, artificial lift optimization, and flow assurance strategies. The ability to detect subtle variations in flow behavior enhances operational responsiveness and supports proactive production optimization.

Moreover, dependable allocation data underpins robust reservoir surveillance and modeling. It enhances the accuracy of history matching, reservoir simulation inputs, and material balance calculations—ultimately leading to more reliable forecasting, better-informed field development planning, and optimized recovery strategies.

This paper is among the few that quantitatively link field measurement uncertainty to system-level impacts on hydrocarbon allocation. It provides practical insights into how transitioning from allocation-grade to custody-grade meters can significantly enhance the accuracy of hydrocarbon accounting. Furthermore, it introduces a replicable framework for evaluating value-at-risk stemming from measurement errors in multi-stakeholder oil and gas operations.

To maximize the benefits of improved measurement integrity, the following actions are recommended:

- **Prioritize upgrades for high-volume streams** where financial exposure is most significant.
- **Conduct comprehensive uncertainty analyses** to identify and quantify potential allocation discrepancies.
- **Implement real-time monitoring tools** to track and manage measurement performance.
- **Align metering practices with commercial and regulatory requirements** to ensure transparency and stakeholder confidence.

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valued experience. The authors would also like to thank ADNOC Onshore Management for approving the paper.

Nomenclature

API MPMS	American Petroleum Institute Manual of Petroleum Standards
BS&W	Basic Sediment and Water
CDS	Central Degassing Station
EPC	Engineering, Procurement, and Construction
FSGPR	Field Sustainable Gas Production Rate
FSOPR	Field Sustainable Oil Production Rate
NIST	National Institute of Standards and Technology
NPV	Net present value
PMAS	Production Measurement and Allocation System
RF	Field Reconciliation Factor
ROI	Return on Investment
SVP	Small Volume Provers
WRF	Well Reconciliation Factor

References

1. ADNOC Onshore Production Measurement Manual
2. ADNOC Production Back Allocation Guideline
3. API MPMS Chapters 4 Calibration and Traceability Standards
4. API MPMS Chapter 8.1 and ASTM D4057 For Manual Sampling
5. API MPMS Chapter 8.2 For Automatic Sampling
6. API MPMS Chapter 12.2.2 and 12.2.3.
7. ISO 5168 and API MPMS Chapter 13.3, Measurement of fluid flow - Procedures for the evaluation of uncertainties.